

Project management offsite drone based construction visibility platform

PURPOSE

To provide construction teams with real-time ground truth by automating :

- Site progress tracking
- 3D visualization of structures
- Drone-based LiDAR scanning
- Accurate measurements & reporting

The platform reduces delays, rework, manual inspection errors, and improves decision-making.

WHAT ARE WE SOLVING?



Limited visibility
into site progress



Safety risks due
to inconsistent
inspections



Manual measurement
errors



Slow reporting
and communication
gaps



Cost overruns from
delays & rework



Fragmented data
across teams

KEY FEATURES

- Drone & LiDAR Data Capture
- Automated aerial missions to scan construction sites, capture 4K imagery & LiDAR point clouds.
- 3D Visualization Dashboard
- Real-time site digital twin using CesiumJS / Three.js.
- Smart Measurements
- Instant distance, area & volume calculations for accurate progress tracking.
- Real-Time Alerts & Notifications
- Secure Cloud Processing

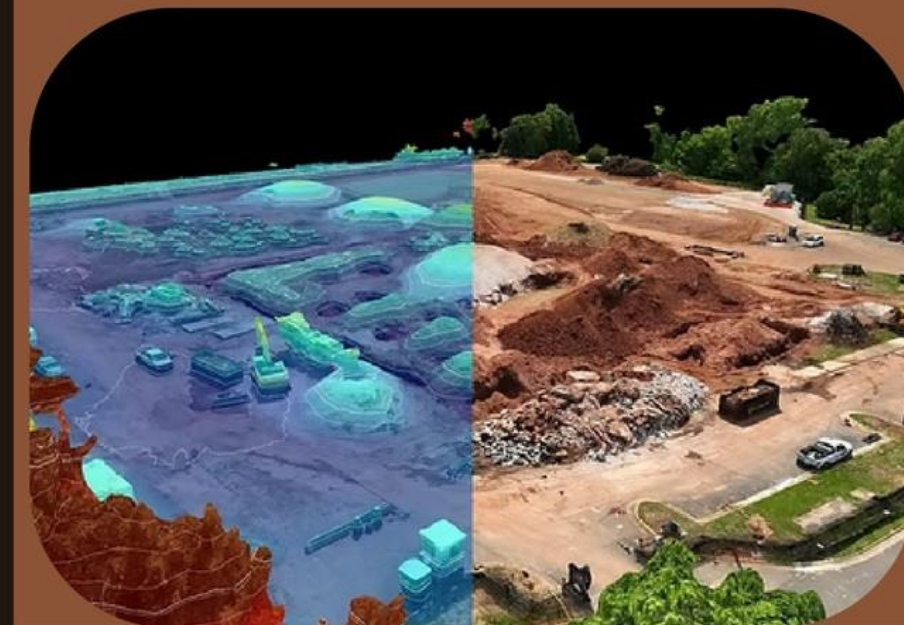
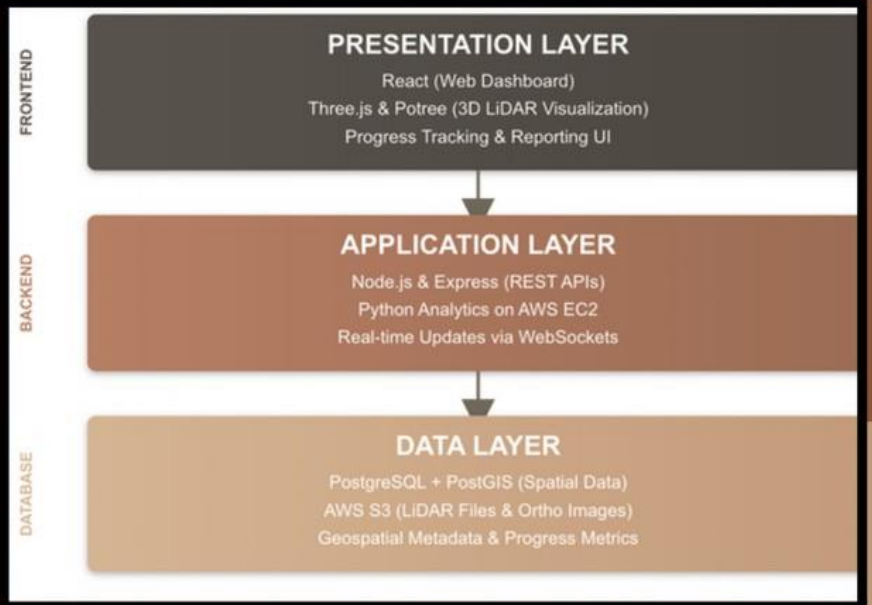
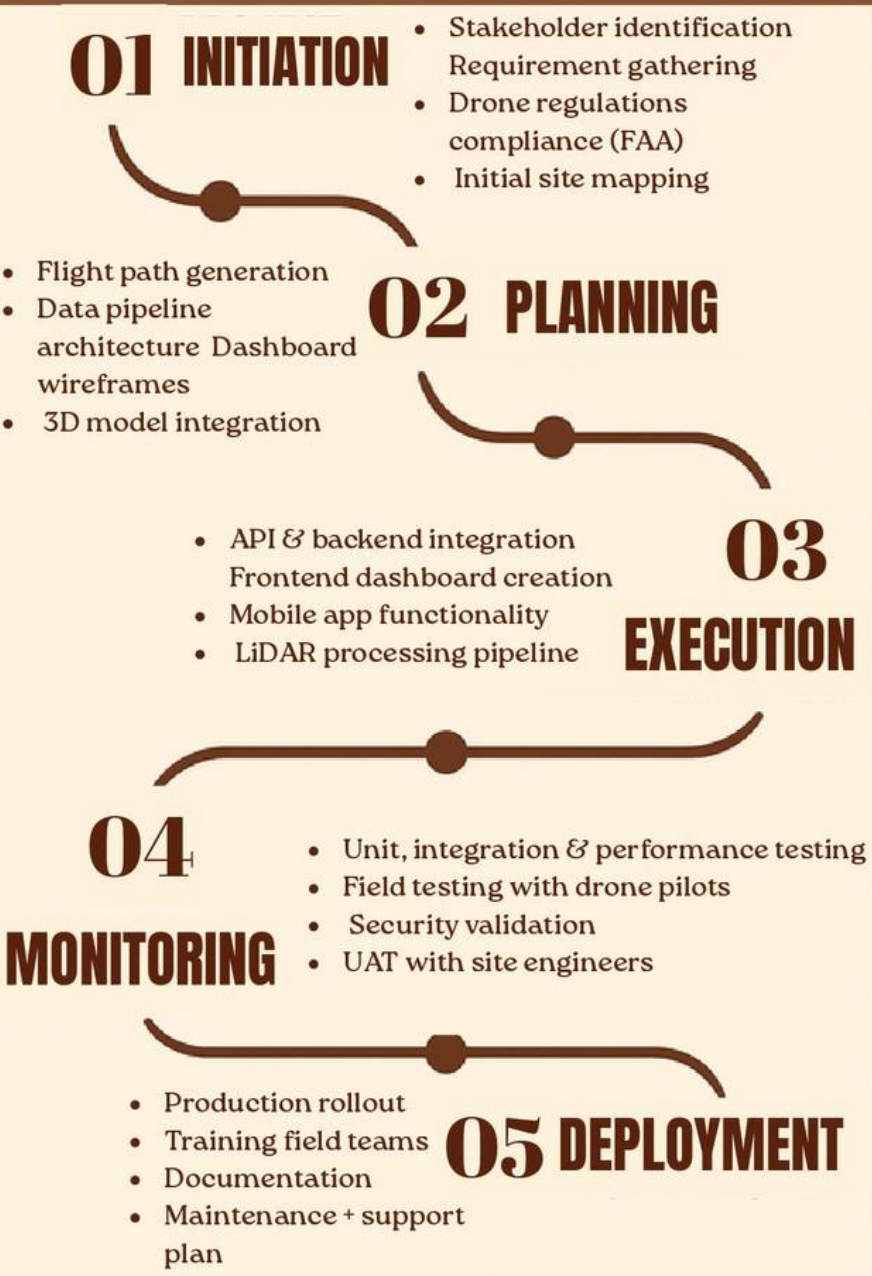


fig : Mapping and 3D modelling of a construction site

PROJECT ROADMAP



BENEFITS

- 01** Increased accuracy in construction planning
- 02** Consistent real-time communication
- 03** Higher worker safety
- 04** Transparent stakeholder visibility

TECHNOLOGY STACK

Drone Hardware: DJI Matrice / Phantom Series

LiDAR Sensors: RIEGL / Livox

Backend: Python, Node.js, Flask

Frontend: React, Flutter

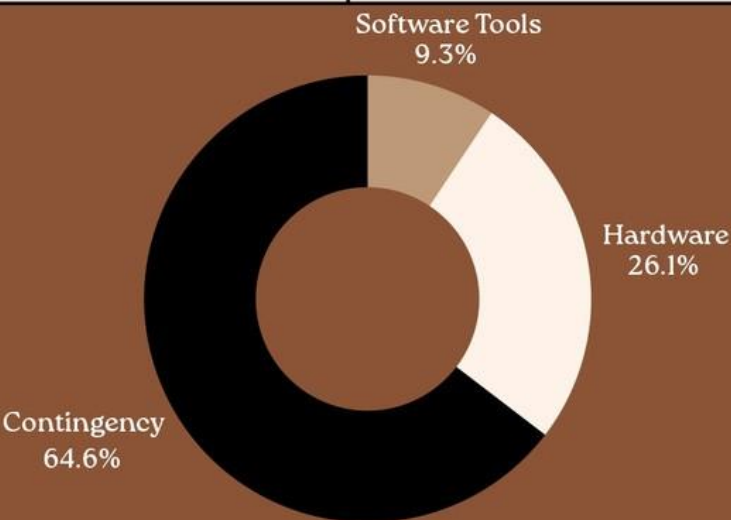
Cloud: AWS EC2, S3, Lambda, RDS (PostGIS)

Visualization: CesiumJS, Three.js

CI/CD: GitHub Actions, Docker

FINANCIAL PLAN

Resources	Total Cost
Labor	\$2,974,400
Software Tools	\$90,272
Hardware	\$251,698
Contingency	\$624,023
Total	\$3,940,393



RESOURCE ALLOCATION

ROLE INVOLVEMENT					
	P1	P2	P3	P4	P5
Project Manager	●●●	●●●	●●●●	●●●	●●●
Developers	●●●	●●●●	●●●●	●●●●	●●●
UI/UX Designer	●●●	●●●●	●●●	●●●	●●
Data Engineer	●●	●●●	●●●	●●	●●
QA Team	●	●●	●●●●	●●●●	●●
QA Automation	●	●●	●●●●	●●●	●●
DevOps	●	●●	●●●●	●●●	●●
Security Architect	●	●●●	●●●	●●	●●
Technical Writer	●	●●●	●●●	●●●	●●
Scrum Master	●	●●●	●●●●	●●●	●●

Key: ● = Low involvement ●● = Medium ●●● = High ●●●● = Very High ●●●●● = Very High